

SAT Hard Question 10

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Novel Prep



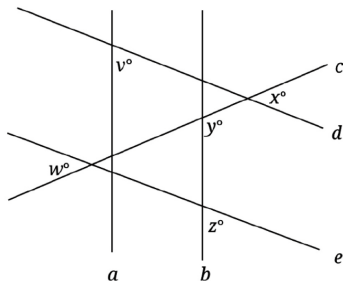
Novel Prep

Excellence in SAT Prep

Parallel Lines and Transversals – Angles

In the figure, parallel lines a and b are intersected by lines c , d , and e .

If $z = 49^\circ$, $y = 136^\circ$, and $v < z$, which statement about x and w must be true?



- A. $x < w$
- B. $x > w$
- C. $x = w$
- D. $x + w = 90$

Solution: Parallel Lines and Angles

Given:

- $z = 49^\circ$
- $y = 136^\circ$
- $v < z$

Step 1: Solve for x :

$$x = 180^\circ - y = 180^\circ - 136^\circ = 44^\circ$$

Step 2: Understand the relation between v and w :

- w corresponds to v via alternate interior angles
- Since $v < z = 49^\circ$, then $w < 49^\circ$

Step 3: Compare:

$$x = 44^\circ, \quad w < 49^\circ \Rightarrow \boxed{x > w}$$

Correct Answer: **B. $x > w$**

Minimum Value of Exponential Functions

The functions f and g are defined by the given equations below, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines?

- I. $f(x) = 16(1.3)^{x+5}$
 - II. $g(x) = 16(1.09)(1.3)^{x+3}$
- A. I only
 - B. II only
 - C. I and II
 - D. Neither I nor II

Solution: Minimum Value as Constant or Coefficient

Function I: $f(x) = 16(1.3)^{x+5}$

- Since $1.3 > 1$, the exponential function increases as x increases.
- So $f(x)$ is an increasing function.
- The minimum value occurs at $x = 0$:

$$f(0) = 16(1.3)^5 \approx 16(3.7129) \approx 59.41$$

- The minimum value is not 16, and 16 is not the minimum value of $f(x)$.

Function II: $g(x) = 16(1.09)(1.3)^{x+3}$

- $(1.3)^{x+3}$ increases as x increases, and 1.09 is a constant.
- So $g(x)$ is also increasing.
- The minimum value occurs at $x = 0$:

$$g(0) = 16(1.09)(1.3)^3 \approx 16(1.09)(2.197) \approx 38.41$$

- Again, 16 is not the minimum value, and is not part of the output.

Conclusion: Neither function displays its minimum value as a constant or coefficient.

Correct Answer: D. Neither I nor II

Y-intercept of a Rational Function

The table shows values of x and corresponding values of $g(x)$, where:

$$g(x) = \frac{f(x)}{x+3} \quad \text{and} \quad f(x) \text{ is a linear function}$$

What is the y -intercept of the graph of $y = f(x)$ in the xy -plane?

x	$g(x)$
-15	3
-6	0
9	5

- A. (0, -6)
- B. (0, 4)
- C. (0, 8)
- D. (0, 24)

Solution: Finding the Y-intercept of $f(x)$

Step 1: Use the formula:

$$g(x) = \frac{f(x)}{x+3} \Rightarrow f(x) = g(x)(x+3)$$

Step 2: Calculate $f(x)$ values:

- At $x = -15$, $f(x) = 3(-15 + 3) = 3(-12) = -36$
- At $x = -6$, $f(x) = 0(-6 + 3) = 0$
- At $x = 9$, $f(x) = 5(9 + 3) = 5(12) = 60$

Step 3: Use two points to find the line $f(x)$:

- Points: $(-6, 0)$ and $(9, 60)$
- Slope $m = \frac{60-0}{9-(-6)} = \frac{60}{15} = 4$
- Line: $f(x) = 4x + b$, plug in $(-6, 0)$:

$$0 = 4(-6) + b \Rightarrow b = 24$$

Y-intercept: $(0, 24)$

Correct Answer: D. $(0, 24)$

Percent Problem – Variables and Percentages

The positive number a is 2106% of the sum of the positive numbers b and c , and b is 81% of c .

What percent of b is a ?

- A. 4706%
- B. 2600%
- C. 38.12%
- D. 21.87%

Solution: Percent Relationship

Step 1: Express b in terms of c :

$$b = 0.81c$$

Step 2: Use the information that a is 2106% of the sum of $b + c$:

$$a = 2106\% \cdot (b + c) = 21.06 \cdot (0.81c + c) = 21.06 \cdot 1.81c = 38.065c$$

Step 3: What percent of b is a ?

$$\frac{a}{b} = \frac{38.065c}{0.81c} = \frac{38.065}{0.81} \approx 47.006 = 4706\%$$

Correct Answer: A. 4706%

Polynomial Roots – Sum of Roots

In the given equation, u is a positive constant. The sum of the solutions to the equation is 1. What is the value of u ?

$$7x^2(3x - 11)(3x - u) = 0$$

- A. 1
- B. -8
- C. $\frac{11}{3}$
- D. $\frac{8}{3}$

Solution: Sum of Solutions

Step 1: Identify the roots from the factored form:

$$7x^2(3x - 11)(3x - u) = 0 \Rightarrow x = 0 \text{ (double root)}, \frac{11}{3}, \frac{u}{3}$$

Step 2: Use sum of all roots:

$$0 + 0 + \frac{11}{3} + \frac{u}{3} = 1 \Rightarrow \frac{11 + u}{3} = 1 \Rightarrow 11 + u = 3 \Rightarrow u = 1 - 11 = -8$$

Solving a Quadratic Equation

Which of the following values of x satisfy the equation

$$-3x^2 + 12x + 4 = 0?$$

- A. 0
- B. -2 and 2
- C. $2 - \frac{2\sqrt{6}}{3}$ and $2 + \frac{2\sqrt{6}}{3}$
- D. $2 - \frac{4\sqrt{3}}{3}$ and $2 + \frac{4\sqrt{3}}{3}$

Solution: Solving a Quadratic Equation

Step 1: Identify coefficients.

• $a = -3, b = 12, c = 4$

Step 2: Use the quadratic formula.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Step 3: Compute discriminant.

$$\Delta = b^2 - 4ac = 144 + 48 = 192 = 64 \cdot 3 \Rightarrow \sqrt{192} = 8\sqrt{3}$$

Step 4: Solve for roots.

$$x = \frac{-12 \pm 8\sqrt{3}}{-6} = 2 \pm \frac{4\sqrt{3}}{3}$$

Correct Answer: D

Angles Formed by Parallel Lines

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of these eight angles is $(7x - 250)^\circ$. The sum of the measures of four of the eight angles is k .

Which of the following could **not** be equivalent to k for all values of x ?

- A. $-14x + 1540$
- B. $14x - 320$
- C. $-28x + 1720$
- D. 360

Solution: Parallel Line Angle Sums

A is not equivalent to k

Triangle Inequality Theorem

The triangle inequality theorem states that the sum of any two sides of a triangle must be greater than the length of the third side.

If a triangle has side lengths of 7 and 10, which inequality represents the possible lengths, x , of the third side of the triangle?

- A. $x < 17$
- B. $x > 17$
- C. $3 < x < 17$
- D. $x < 3$ or $x > 17$

Solution to Triangle Inequality Problem

Let the three sides be 7, 10, and x . Apply the triangle inequality:

$$x + 7 > 10 \Rightarrow x > 3$$

$$x + 10 > 7 \Rightarrow x > -3 \quad (\text{always true since } x > 3)$$

$$7 + 10 > x \Rightarrow x < 17$$

Combining the valid inequalities:

$$3 < x < 17$$

Correct Answer: C. $3 < x < 17$